

Writing a Good Mathematical Paper

A compact guide for authors and AI assistants

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Help a reader who knows the field state the result, explain why the proof works and use it correctly. Use these instructions to resolve a difficulty, not to add a sentence about every item.

1. Decide what the paper contributes. Choose a coherent principal result and an intended reader. Say what becomes possible or understood because of the result, how it differs from the nearest prior work, and what remains open. For a conditional result, explain how restrictive the hypothesis is. An abstract should identify the actual result and mechanism; theorem count and generality alone do not establish importance.

2. Read the kind of mathematics you are writing. Read nearby papers in the same subject and genre, including complete local arguments. Study terminology, hypothesis placement, notation, proof order and attribution. Also seek a proof with the same explanatory difficulty: a choice, a limit or an estimate. Check the edition and passage. Adapt the reason for a choice; write your own sentences and supply your own mathematical warrant.

3. Check the statement before polishing its prose. Check domains, quantifiers, signs, end-points, constants, parameter dependencies and both directions of any claimed equivalence. State assumptions where the claim is made. Compare titles, abstracts and conclusions with the full statement. Separate witnesses need not be a common witness: over the natural numbers, $\forall n \exists x (x > n)$ does not imply $\exists x \forall n (x > n)$. An exposition pass that exposes a proof gap must refer it for mathematical review; elegant wording cannot close it.

4. Organise by logical dependence. Present the question, the result and the proof's main idea before technical detail accumulates. Use an example where it prepares the reader for an unfamiliar operation. State a lemma's purpose before its proof; place prerequisites before use. Let headings name the object, result or operation. Give a reader entering at a theorem enough information to find its assumptions and proof. Choose length to suit the argument; there is no universal page quota.

5. Explain the hard transition. Name the quantity to control and the accuracy required before deriving an estimate. Explain which constraint selects a parameter or change of variables. For an interval certificate, compare the error with both endpoint gaps. For a small-integer contradiction, justify integrality, nonvanishing and $|N| < 1$ separately. When a technique recurs for a new task, name that task. Spend detail on the decisive inference and make clear what remains to prove.

6. Separate construction from completion. Account for each choice and preserved condition. A classification needs both construction and exhaustiveness. An enclosure alone does not show that every target is attained. For example, $r_n = x - S_n$, $0 \leq r_n \leq \varepsilon_n$ and $\varepsilon_n \rightarrow 0$ give $S_n \rightarrow x$; repeated choice alone does not. State which parameters are fixed and justify the limit of the whole summand, including any growing product. To interchange a limit and a sum, identify an applicable theorem and check its hypotheses.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

7. Make examples and figures do work. Choose a small instance that reveals a definition, invariant, obstruction or necessary hypothesis. Show the operation and its consequence, including an endpoint case when the convention matters. Explain what the example leaves unproved. A finite check supports an infinite claim only through a proved reduction. Check that a witness belongs to the claimed object. Label a diagram's objects, maps and conventions, and keep its explanation nearby.

8. Use the field's words and purposeful notation. Keep established terms when their definitions fit, such as "Stieltjes moment sequence". Replace private labels by the actual condition; $u_n/2^n \rightarrow 0$ is weaker than subexponential growth. Define symbols before use, keep their roles stable and remove names that save no work. Translate borrowed indices, coordinates and normalisations. Distinguish a sum, its partial sums and its remainders, and a value from the interval enclosing it.

9. Make sentences express the argument. Replace "We proceed with the estimate" by a reason: "Orthogonality makes the first sum zero; it remains to bound the second." Make connectives and referents unambiguous. After "similarly", check that the same hypotheses apply. Punctuate displayed equations as sentences. Keep parallel claims parallel and conditions beside their consequences. Remove filler; retain precise terms.

10. Credit the exact input at its use. Distinguish applied theorems, adapted constructions, historical attribution, terminology and expository models. Check the original's hypotheses, version and passage. Say what an adaptation retains and changes. Credit supplied ideas and corrections locally. A writing analogy supplies no proof. Report only reading and checks actually performed.

11. State what has actually been established. Separate logical scope from evidence. A conditional implication can have an ordinary or formal proof; label computations and conjectures separately. Keep unproved inputs beside the dependent result. A checked nearby lemma does not formalise the whole proof; missing formalisation does not invalidate an ordinary proof. Give computations their finite scope and reproduction details in one verification section. For a one-sided test, distinguish what passing proves from what failure leaves undecided. Pointwise eventual detection need not supply a uniform stopping bound or the witnesses a quantified theorem requires.

12. Preserve useful detail and informative failures. When there is a longer companion, keep the short argument readable and omitted detail recoverable. Verify passage destinations and reconcile statements, assumptions and evidence in both directions. Preserve a useful failed route's attempted implication, assumptions, witness and missing ingredient. Distinguish flawed steps, failed methods and counterexamples. Explain the proof without inventing its discovery history. Cite omitted arguments by section or theorem; keep build commands and source identifiers in the verification record.

13. Review changes as claims, then as prose. Compare old and new meanings, including explanatory words beside unchanged formulas. Correct overclaims and stale underclaims. Removing shorthand must preserve its hypotheses; a familiar term must meet its definition. Accept, repair, reject or defer each substantive proposal for a stated reason. Generalise a lesson only as far as its evidence supports, retaining its limit. Keep useful rejected revisions recoverable.

14. Read the delivered paper from a cold start. Render the final sources. Inspect formulas, captions, page breaks, bibliography and cross-reference destinations. Read without supplying steps from memory: can the intended reader reconstruct the hard inference, apply the result in an example and identify an excluded case? Record actual difficulties. A successful build is not a comprehension test. Add prose only to resolve a difficulty, improve accuracy or recover an argument.

Further reading. Halmos, *How to write mathematics*, §§3–5 (1970); Knuth et al., *Mathematical Writing*, §§1–3 (1988); Gowers, *Examples first!* (2007); Tao, *On writing*. Worked cases, sources and revision history: the *long companion*.